Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/04/2025
PI Name:	Galazka, Jonathan Ph.D.		
Project Title:	Responses of Microbes and Microbial Communities to Prolonged Exposure to Space Radiation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	605-604-3950
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Research Division		
PI Address 2:	MS288-2		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	03/01/2022	**End Date:**	02/28/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date has changed to 02/28/2025 per F. Hernandez/ARC (Ed., 6/18/24) NOTE: End date has changed to 02/01/2025 per F. Hernandez/ARC (Ed., 1/16/24) NOTE: End date has changed to 03/01/2024 per F. Hernandez/ARC (Ed., 3/30/23)		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):	Barrick, Jeffrey Ph.D. (University of Texas, Austin)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The built environment of spaceships is host to a microbial community that affects crew and craft alike. While the static composition of this community has been characterized, and its temporal dynamics examined, the mechanisms controlling its make-up and evolutionary trajectory are not understood. Systematic analyses of microbial diversity such as the "Earth Microbiome Project" and the "Human Microbiome Project" have shown consistent patterns in community composition and function. Understanding the ecological origins of these patterns remains a major challenge, as it requires connecting processes that occur at varying temporal and spatial scales. However, it is clear that the state and trajectories of microbial communities are in-part determined by their physical environments. In this regard, the spaceflight environment includes numerous interacting factors that differentiates it from Earth environments, including an altered atmospheric composition, reduced gravity (and thus altered fluid dynamics), and increased ionizing radiation. These factors impart selective pressures on microbial communities that effect their evolutionary trajectories and thus, the risks and benefits these communities represent to crew and craft. The radiation environment of space leads to chronic exposure to low doses (<0.1 Gy/hr) and is difficult to mimic on Earth. Thus, little is known about how microbial communities in spacecraft will respond and evolve. Therefore, given the limitations of existing studies, we propose to empirically determine how exposure to low doses of ionizing radiation for thousands of cell divisions affects rates of mutation accumulation in microbes and the trajectory of microbial evolution. In this way, we will provide a critical set of data for the design of safe and robust space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research on the impacts of chronic low dose rate irradiation on microbial mutation rates and evolution will impact our understanding of how terrestrial microbes respond to analogous environments. This includes naturally occurring environments and those impacted by human activities. Moreover, our research will help understand how human associated microbes and microbiomes respond to radiotherapies.
Task Progress:	Optimized “fluctuation assay” with E. coli strain REL1206 was completed. This assay allows for an estimate of the genetic mutation rate in an environmental and genetic background without reliance upon sequencing. This will be performed in standard radiation environment (terrestrial background) and with exposure to modeled space radiation within leaded enclosure to ensure mutation rates are as previously reported in terrestrial background and that they are elevated. Partial development of quantitative model of radiation environment within the E. coli REL1206 cultures. This model allows for an estimate of the number of times an E. coli cell would be expected to be transited by a gamma photon from the 57Co plates. This model will be used to better contextualize sequencing results.
Bibliography Type:	Description: (Last Updated:)